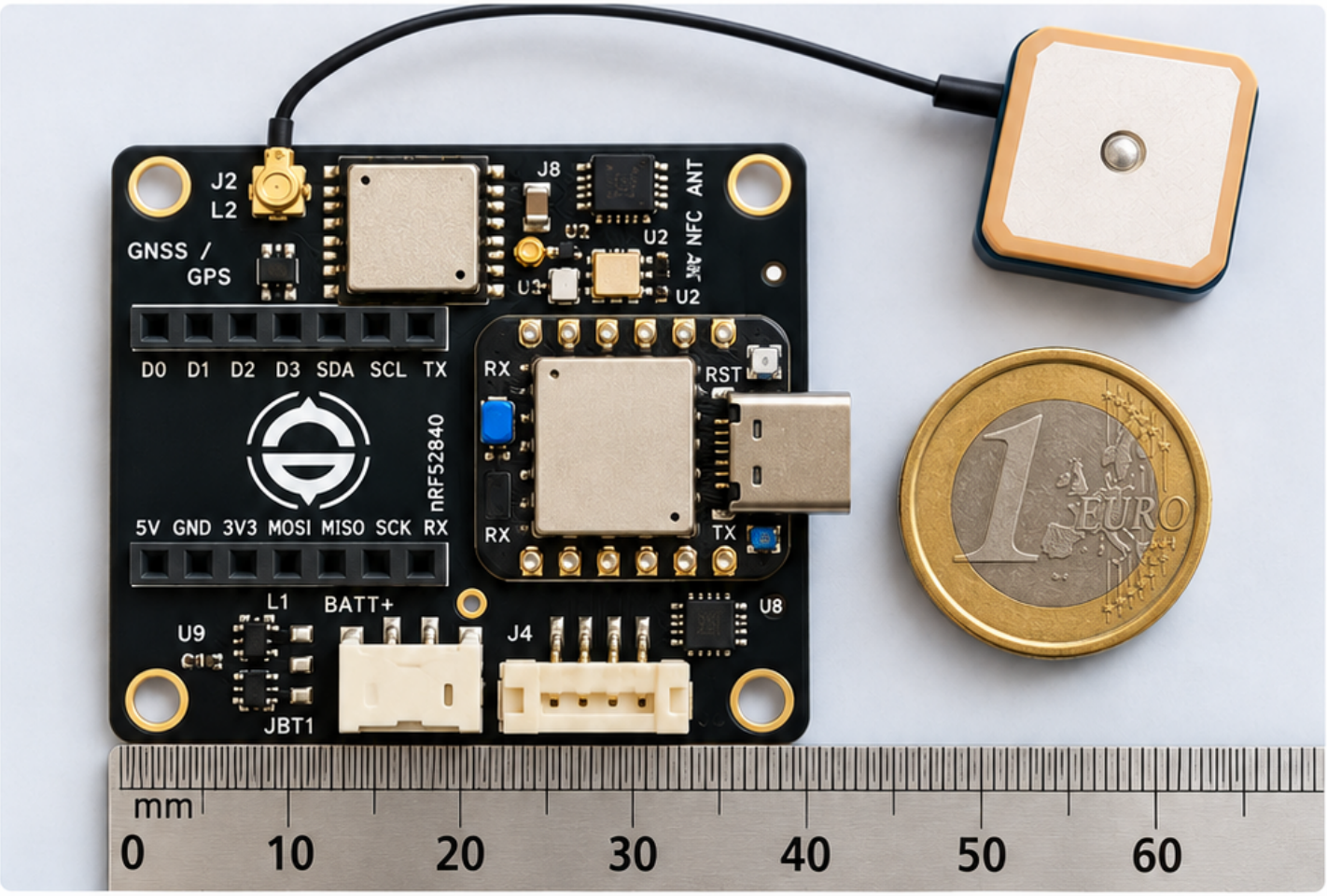
EMBEDDED SYSTEMS PORTFOLIO

Nordic BLE Hardware & Firmware Engineering

Hardware Design • Embedded Firmware • Wireless • Manufacturing • Validation

End-to-end development of connected embedded systems - from circuit architecture and PCB design through Nordic firmware, wireless connectivity, production handoff and physical validation.



GNSS
External active antenna
Positioning / RF

NFC
External loop antenna
Tap / proximity interface

LiPo POWER
1-cell battery input
Portable low-power operation

COMPLETE EMBEDDED ENGINEERING CAPABILITY

SCHEMATIC + PCB
Design & implementation

ZEPHYR / NCS
Embedded firmware

BLE + RF
Custom GATT & wireless

BOM / CPL / GERBER
Manufacturing handoff

PHYSICAL VALIDATION
Real hardware testing

nRF52840 | KiCad | Embedded C | Zephyr RTOS | Nordic nRF Connect SDK | BLE GATT | Sensors | GNSS / RF

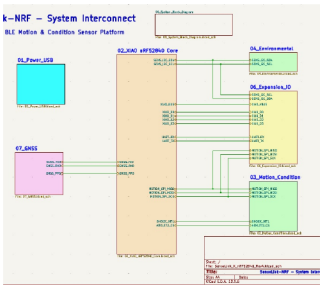
Engineering evidence and detailed project work follow.

NORDIC BLE HARDWARE & FIRMWARE ENGINEERING

From System Concept to Working Connected Hardware

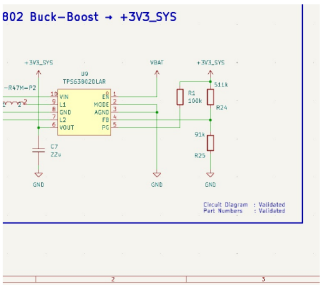
Hardware architecture, schematic design, PCB implementation, manufacturing handoff, embedded firmware and physical system validation.

IDEA → ARCHITECTURE → SCHEMATIC → PCB → MANUFACTURING → FIRMWARE → VALIDATION → FINAL HARDWARE



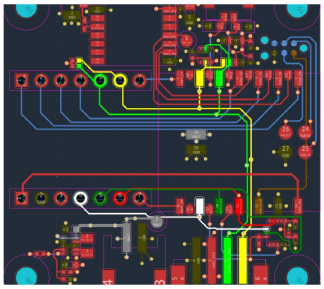
SYSTEM ARCHITECTURE

Requirements translated into defined system blocks and interfaces.



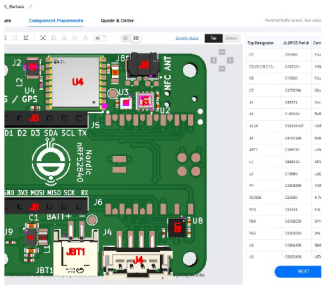
SCHEMATIC DESIGN

Power, sensing, RF/GNSS and debug circuits developed hierarchically.



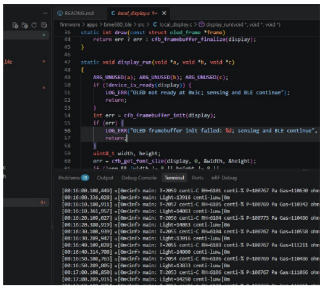
PCB IMPLEMENTATION

Placement, routing and bring-up access implemented in KiCad.



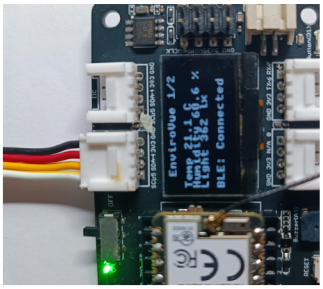
MANUFACTURING

Gerbers, BOM and CPL taken through a real assembly workflow.



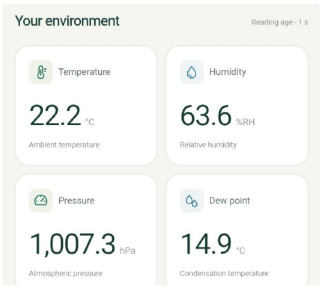
EMBEDDED FIRMWARE

Modular Embedded C on Nordic hardware using Zephyr and NCS.



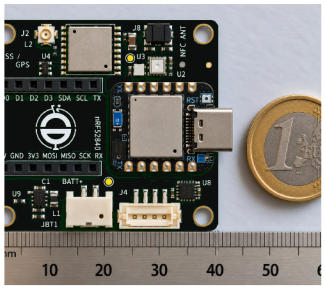
PHYSICAL VALIDATION

Sensors and firmware exercised on physical Nordic hardware.



LIVE APPLICATION

EnviroVue closes the data chain with live telemetry and device control.



COMPACT HARDWARE

Final hardware shown against a 1 Euro coin and millimetre scale.

END-TO-END ENGINEERING

CONCEPT -> ELECTRONICS -> PCB -> MANUFACTURING -> FIRMWARE -> VALIDATION -> HARDWARE

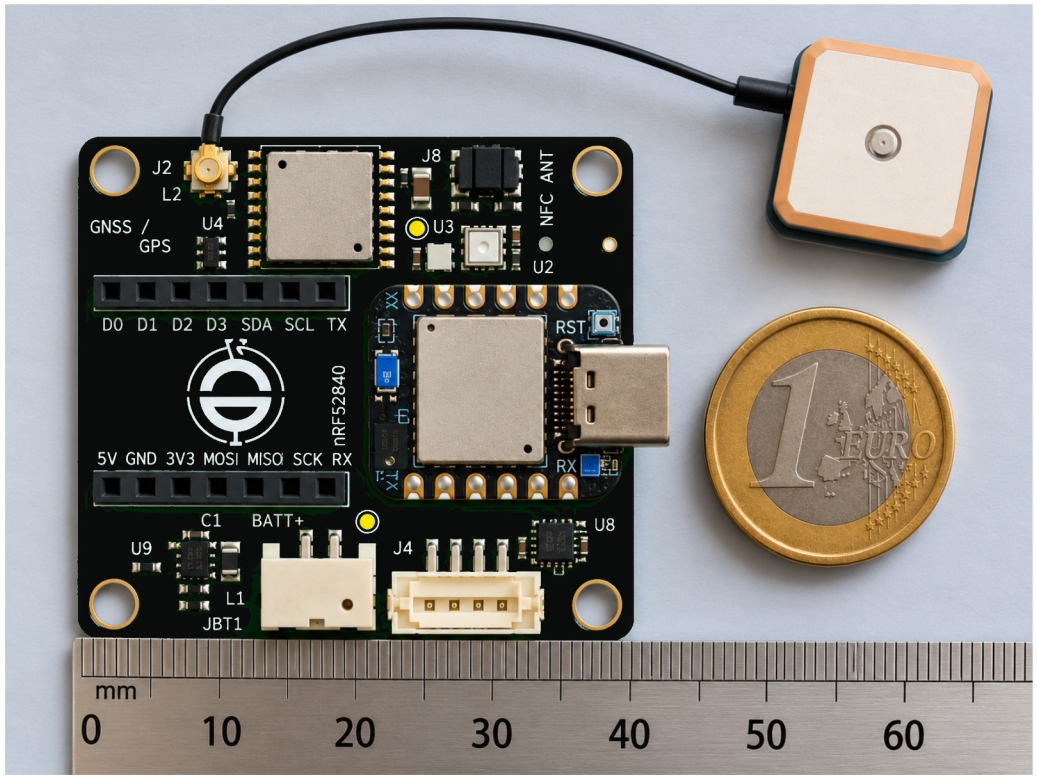
nRF52840 | KiCad | Embedded C | Zephyr RTOS | Nordic nRF Connect SDK | I2C / SPI / UART | BLE | GNSS / RF

Engineering process overview - detailed evidence is provided throughout the portfolio.

CUSTOM nRF52840 HARDWARE DESIGN

Feature-Rich Nordic BLE Sensor & GNSS Hardware

Custom KiCad design combining wireless, sensing, positioning, modular expansion and low-power system architecture.



ENGINEERING FEATURE SET

BLE + NFC + DEBUG

nRF52840 BLE core with dedicated NFC1/NFC2 connections and SWD programming/debug.

POWER ARCHITECTURE

1-cell LiPo system with TPS63802 buck-boost generating the +3V3_SYS rail, plus power test access.

MOTION / CONDITION

ADXL372 high-g motion and condition sensing over SPI with interrupt signalling.

ENVIRONMENTAL + LIGHT

BME688 environmental sensing plus VEML6035 ambient-light sensing on I2C.

GNSS + ACTIVE ANTENNA

MAX-M10S GNSS with UART/PPS, external RF connection and VCC_RF active-antenna bias.

MODULAR EXPANSION

I2C, SPI, UART and GPIO support external sensors, displays, communications and additional wireless interfaces, including LoRa-class radios. Dedicated +3V3_SYS expansion supply.

KICAD / DFT

Hierarchical design: Power/USB -> Core -> Motion -> Environmental -> Expansion -> GNSS. SWD, test access and manufacturing-oriented design checks.

RF / COMPLIANCE: Pre-certified nRF52840 radio module (FCC / CE) provides a strong compliance foundation; final product compliance remains application dependent.

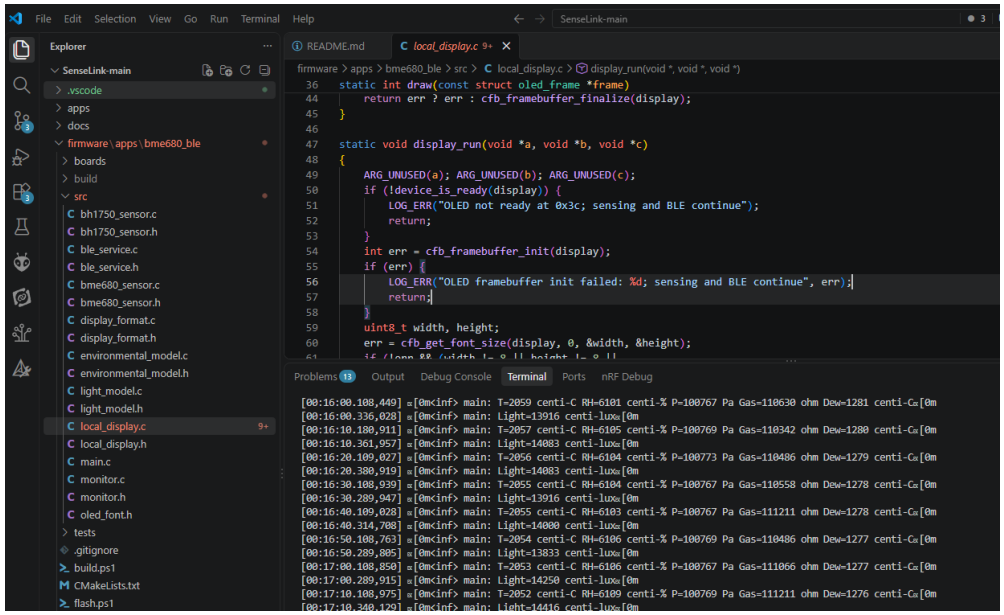
nRF52840 BLE + NFC | LiPo + Buck-Boost | ADXL372 | BME688 + VEML6035 | MAX-M10S + Active Antenna | Modular I2C / SPI / UART / GPIO | SWD / DFT

Custom hardware-design evidence: architecture, schematic capture, PCB implementation, test access and manufacturing preparation.

NORDIC FIRMWARE ENGINEERING

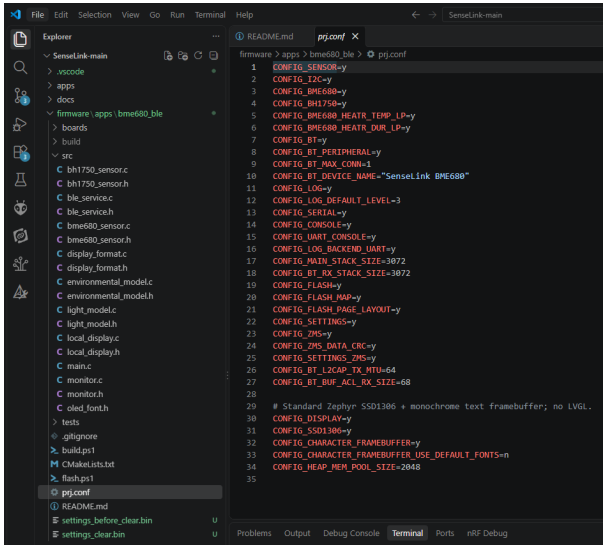
Zephyr RTOS • nRF Connect SDK • Real-Hardware Validation

Modular embedded C firmware developed and exercised on physical Nordic hardware with live sensor telemetry and Bluetooth LE.



MODULAR SOURCE + LIVE TELEMETRY

Separate sensor, BLE, display and application modules with live environmental readings from physical hardware.



ZEPHYR CONFIGURATION

Sensors • I²C • BLE peripheral • logging • flash/settings • SSD1306

MODULAR EMBEDDED C

Dedicated modules for BME680, BH1750, BLE services, environmental/light models, display formatting, monitoring and application control.

ZEPHYR / NCS

Zephyr sensor and I²C frameworks, Bluetooth LE peripheral configuration, UART logging, display support and explicit memory/transport configuration.

PERSISTENT SETTINGS

Flash map/page layout with Zephyr Settings and ZMS storage, including data CRC support for non-volatile application configuration.

PHYSICAL VALIDATION

Firmware built, flashed and exercised on real Nordic hardware; runtime logs show live temperature, humidity, pressure, gas and light telemetry.

SENSORS → MODULAR DRIVERS → APPLICATION LOGIC → BLE → LIVE TELEMETRY

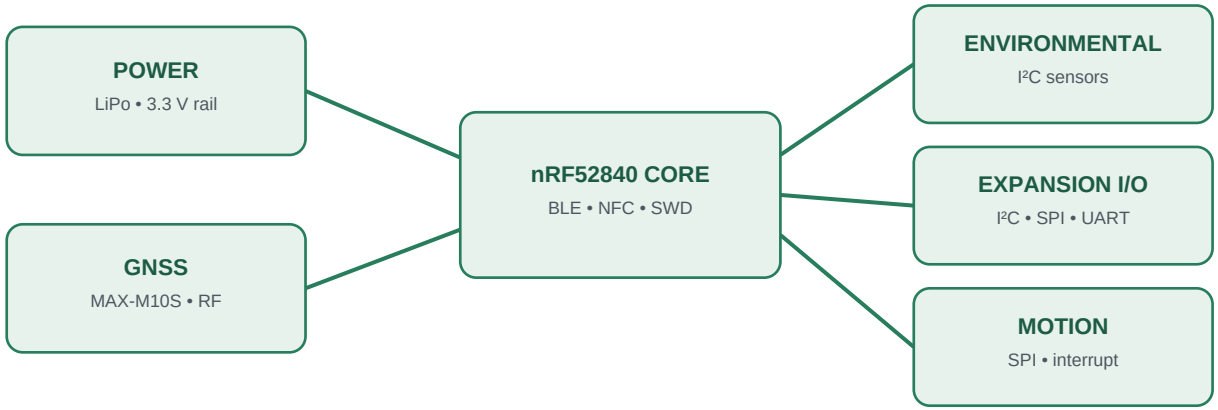
Embedded C | Zephyr RTOS | Nordic nRF Connect SDK | I²C | BLE | Logging | Flash / Settings | Hardware Debug

SCHEMATIC & ELECTRONIC DESIGN

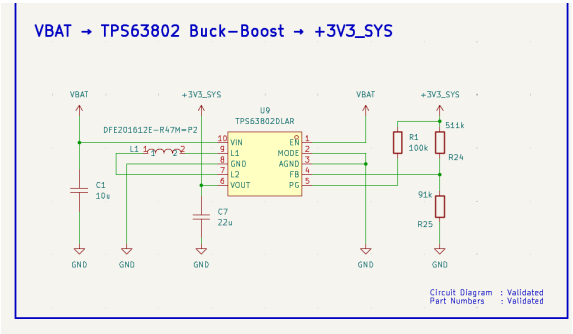
System Architecture • Power • GNSS / RF

Hierarchical circuit design translating system requirements into defined power, sensing, positioning and interface subsystems.

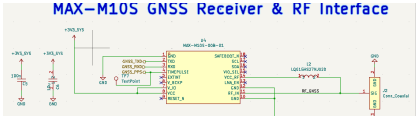
SYSTEM-LEVEL SCHEMATIC ARCHITECTURE



Functional partitioning keeps power, sensors, positioning and external interfaces explicit and reviewable.



POWER ARCHITECTURE



GNSS / ACTIVE-ANTENNA RF INTERFACE

SYSTEM DECOMPOSITION

Hierarchical design separates power, Nordic core, motion, environmental sensing, expansion I/O and GNSS into functional blocks with defined interfaces.

POWER DESIGN

Single-cell battery input is converted to the 3.3 V system rail using a TPS63802 buck-boost stage with external inductor, input/output capacitance and feedback network.

GNSS / RF

MAX-M10S positioning interface includes UART and time-pulse signals plus an external coaxial RF connection. VCC_RF feeds the antenna line through a 27 nH RF choke for active-antenna bias.

DESIGN FOR DEBUG

Dedicated programming/debug access and exposed subsystem interfaces support board bring-up, firmware development and engineering verification.

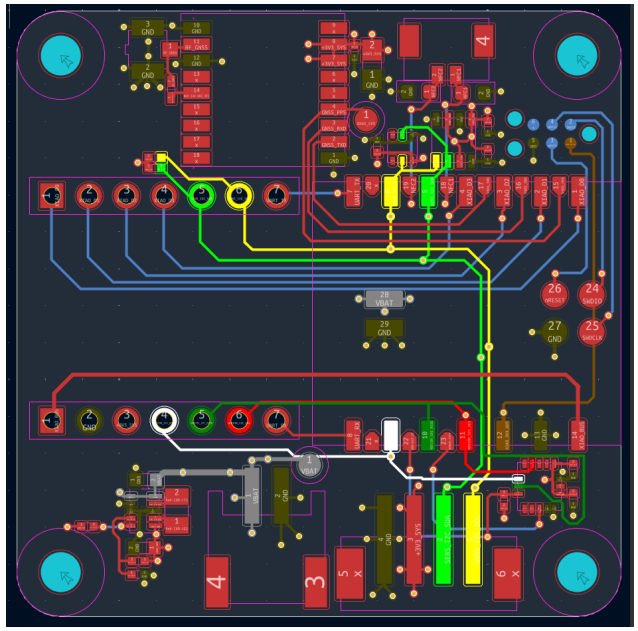
REQUIREMENTS -> SYSTEM ARCHITECTURE -> CIRCUIT DESIGN -> INTERFACE DEFINITION -> PCB IMPLEMENTATION

KiCad | nRF52840 | Power Architecture | I2C / SPI / UART | GNSS / RF | SWD / Debug | Modular Expansion

PCB LAYOUT & MANUFACTURING ENGINEERING

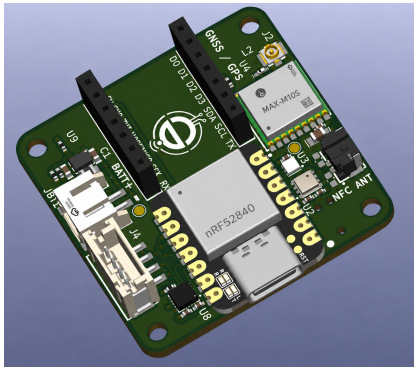
Routed PCB • Placement • DFT • Manufacturing Handoff

Custom Nordic PCB implementation in KiCad, carrying the schematic architecture through placement, routing and fabrication-ready outputs.



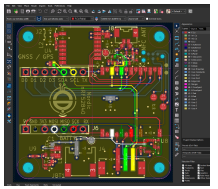
ROUTED PCB - COPPER, VIAS & INTERCONNECT

Clean routing view exposes the implemented signal paths, connector fan-out, debug access and subsystem placement.



3D IMPLEMENTATION

Populated-board integration with nRF52840 core and MAX-M10S GNSS module.



KICAD PCB EDITOR WORKFLOW

PLACEMENT & PARTITIONING

Functional areas are physically organized around the Nordic core, GNSS/RF section, power conversion, sensor interfaces, connectors and expansion access.

ROUTING & FAN-OUT

Implemented copper shows signal breakout from the core to external I/O, GNSS, debug and peripheral interfaces, with vias and layer transitions visible in the routed design.

BRING-UP / DFT

Dedicated SWD/reset access, exposed rails and accessible connectors support programming, board bring-up, firmware development and bench verification.

FABRICATION HANDOFF

The KiCad design is carried through to manufacturing-oriented outputs including BOM, component-position data and Gerber/drill fabrication files.

SCHEMATIC -> PLACEMENT -> ROUTING -> DFT / BRING-UP ACCESS -> BOM / CPL / GERBERS

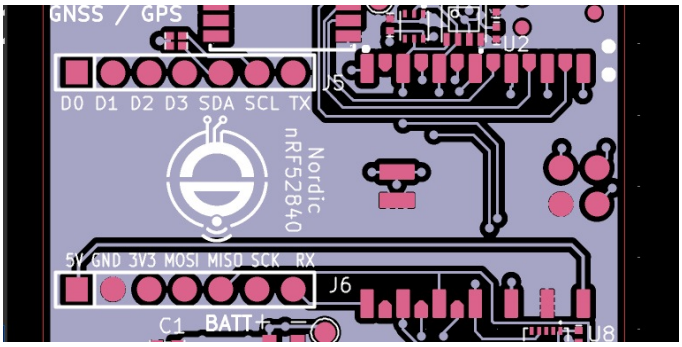
KiCad PCB Editor | Mixed-Signal Layout | GNSS / RF Integration | SWD / Debug | Manufacturing Outputs

MANUFACTURING & PRODUCTION HANDOFF

Assembly-Ready Manufacturing Data

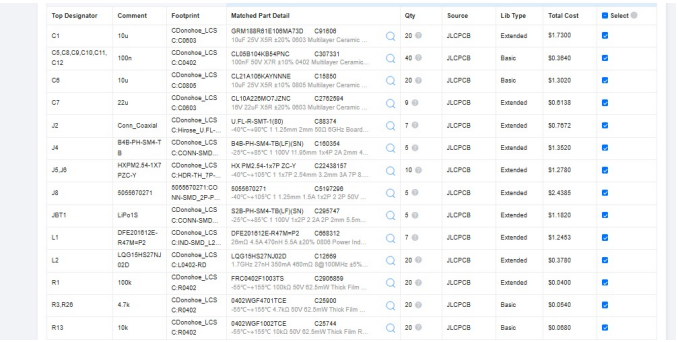
Gerbers • BOM • CPL • Assembly Review • Production Handoff

PCB DESIGN -> GERBER / DRILL -> BOM -> CPL -> ASSEMBLY REVIEW -> MANUFACTURING PACK



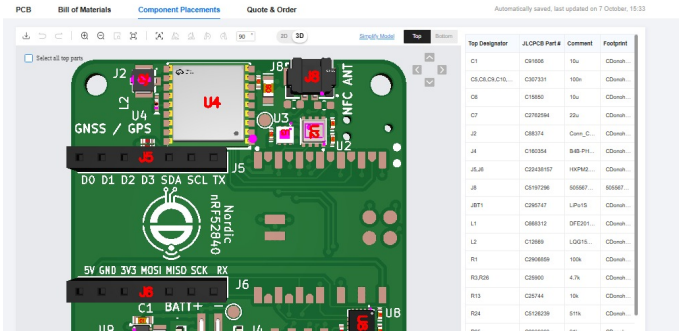
FABRICATION DATA

Gerber Viewer confirms fabrication layers and drill/pad data.



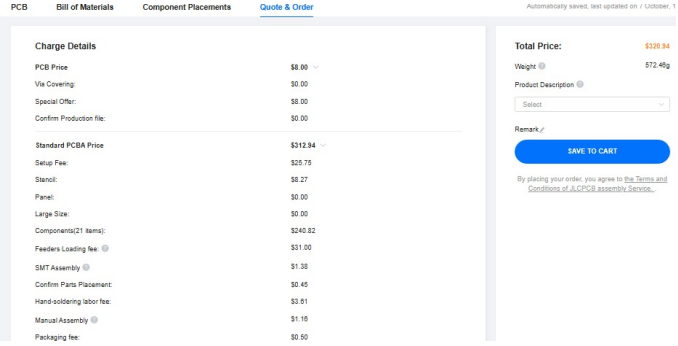
BOM & SOURCING VALIDATION

JLCPCB workflow maps designators, footprints and sourced parts.



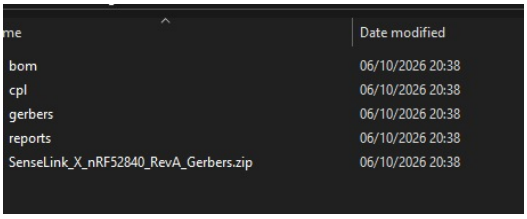
PLACEMENT / CPL REVIEW

JLCPCB interprets pick-and-place data in the assembly preview.



ASSEMBLY WORKFLOW

JLCPCB PCB + SMT data accepted through quotation stage.



ORGANIZED MANUFACTURING DELIVERABLE

Structured release pack separates BOM, CPL, Gerbers and reports, with a packaged Gerber archive ready for fabrication and assembly handoff.

BOM | CPL | GERBERS | REPORTS | FABRICATION ZIP

DESIGN -> FABRICATION DATA -> COMPONENT SOURCING -> PLACEMENT -> ASSEMBLY HANDOFF

KiCad | Gerber / Drill | BOM | CPL / Pick & Place | Component Sourcing | SMT Assembly

BLUETOOTH LE ENGINEERING

Custom GATT • Notifications • Device Configuration

Custom Zephyr/NCS Bluetooth LE interface carrying live environmental data and writable device configuration.

```c
CCC_CALLBACK(humidity)
CCC_CALLBACK(pressure)
CCC_CALLBACK(gas)
CCC_CALLBACK(dew)
CCC_CALLBACK(record)
CCC_CALLBACK(light)

#define MEASUREMENT(uuid, index, callback) \
    BT_GATT_CHARACTERISTIC(uuid, BT_GATT_CHRC_READ | BT_GATT_CHRC_NOTIFY, \
        BT_GATT_PERM_READ, read_measurement, NULL, (void *)&measurement_indices[index]), \
    BT_GATT_CCC(callback, BT_GATT_PERM_READ | BT_GATT_PERM_WRITE)

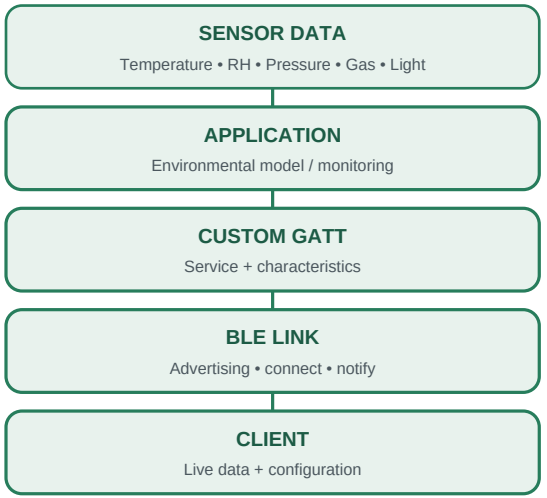
BT_GATT_SERVICE_DEFINE(environmental_service,
    BT_GATT_PRIMARY_SERVICE(&service_uuid),
    MEASUREMENT(&temperature_uuid.uuid, 0, temperature_ccc),
    MEASUREMENT(&humidity_uuid.uuid, 1, humidity_ccc),
    MEASUREMENT(&pressure_uuid.uuid, 2, pressure_ccc),
    MEASUREMENT(&gas_uuid.uuid, 3, gas_ccc),
    MEASUREMENT(&dew_uuid.uuid, 4, dew_ccc),
    BT_GATT_CHARACTERISTIC(&record_uuid.uuid, BT_GATT_CHRC_READ | BT_GATT_CHRC_NOTIFY,
        BT_GATT_PERM_READ, read_record, NULL, NULL),
    BT_GATT_CCC(record_ccc, BT_GATT_PERM_READ | BT_GATT_PERM_WRITE),
    BT_GATT_CHARACTERISTIC(&interval_uuid.uuid, BT_GATT_CHRC_READ | BT_GATT_CHRC_WRITE,
        BT_GATT_PERM_READ | BT_GATT_PERM_WRITE, read_interval, write_interval, NULL),
    BT_GATT_CHARACTERISTIC(&light_uuid.uuid, BT_GATT_CHRC_READ | BT_GATT_CHRC_NOTIFY,
        BT_GATT_PERM_READ, read_light, NULL, NULL),
    BT_GATT_CCC(light_ccc, BT_GATT_PERM_READ | BT_GATT_PERM_WRITE)
);

static const struct bt_data advertising[] = {
    BT_DATA_BYTES(BT_DATA_FLAGS, BT_LE_AD_GENERAL | BT_LE_AD_NO_BREDR),
    BT_DATA_BYTES(BT_DATA_UUID128_ALL, ENV_UUID(0x8f3a0000)),
};
```

CUSTOM GATT SERVICE - ZEPHYR / NCS

Actual implementation: environmental characteristics, READ/NOTIFY, CCC subscriptions, writable interval and advertising.

BLE DATA PATH



CUSTOM GATT

Custom 128-bit service with dedicated environmental characteristics rather than relying only on a standard BLE profile.

READ + NOTIFY

Temperature, humidity, pressure, gas resistance, dew point and light are exposed for client reads and live notification updates.

TWO-WAY CONTROL

A writable interval characteristic lets the BLE client change the device sampling/update interval, with validation handled in firmware.

ROBUST BLE FLOW

Advertising, connection state, CCC subscriptions and notification delivery are integrated into the application architecture.

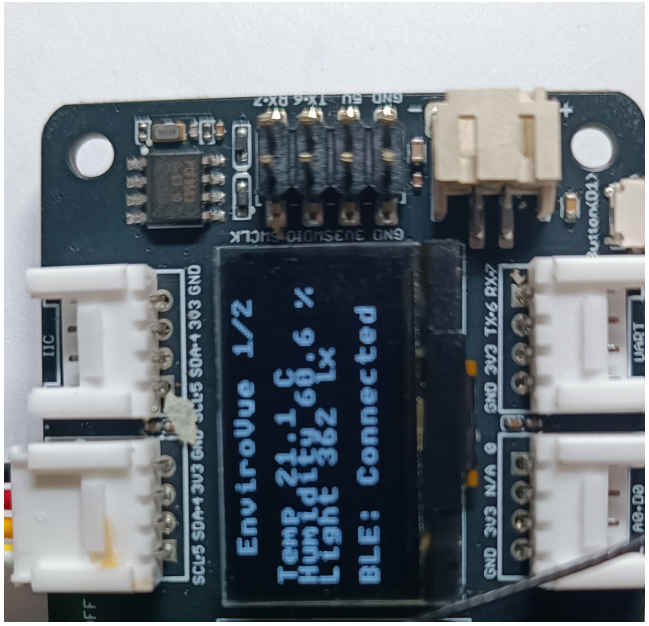
PHYSICAL VALIDATION | Nordic hardware -> Zephyr/NCS -> custom GATT -> BLE client -> live sensor data + device configuration

Embedded C | Zephyr Bluetooth Stack | Custom 128-bit UUIDs | GATT | CCC / Notifications | Advertising

END-TO-END SYSTEM VALIDATION

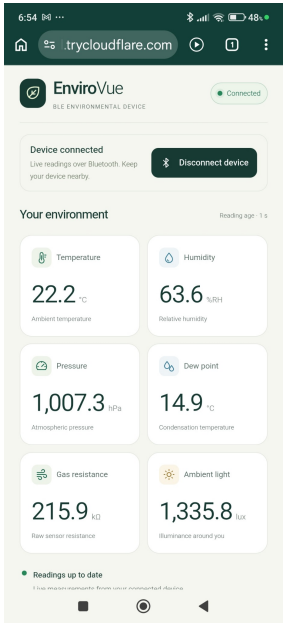
Physical Sensors • Nordic Firmware • BLE • Live Application

Working sensor-to-application chain validated on physical Nordic hardware with live telemetry and bidirectional BLE configuration.



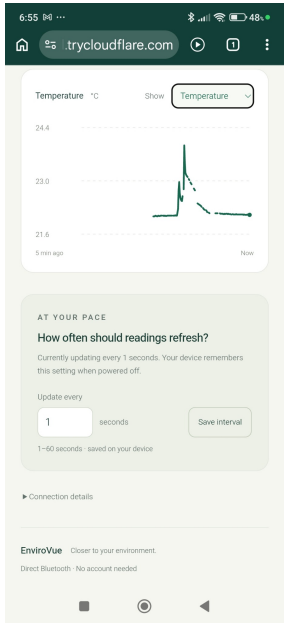
PHYSICAL NORDIC HARDWARE

OLED confirms live environmental values and BLE connection.



ENVIROVUE - LIVE BLE DASHBOARD

Connected client displaying six live environmental measurements.



LIVE TREND + DEVICE CONFIGURATION

Client changes update interval over BLE; device retains the setting.

PHYSICAL VALIDATION

Firmware was built, flashed and exercised on real Nordic hardware rather than simulator-only development.

LIVE SENSOR CHAIN

Environmental and ambient-light measurements are acquired on-device and presented locally before transport over BLE.

BIDIRECTIONAL BLE

Live measurements flow to the client while a writable GATT characteristic allows the application to change the device update interval.

APPLICATION INTEGRATION

EnviroVue demonstrates the complete interface: connection state, live values, recent trends and persistent device configuration.

BME680 + BH1750 -> PHYSICAL NORDIC HARDWARE -> ZEPHYR/NCS -> CUSTOM BLE GATT -> ENVIROVUE

Measured on hardware | Live BLE telemetry | Browser client | Recent-value trends | Writable persistent configuration